

AWS Interview Questions & Answers – Fundamentals (Detailed)

1. What is AWS and what are its key advantages?

Amazon Web Services (AWS) is a comprehensive cloud computing platform provided by Amazon. It offers more than 200 fully featured services including compute, storage, networking, databases, analytics, AI/ML, security, and DevOps tools from global data centers. Key advantages include pay-as-you-go pricing, which eliminates upfront capital investment; global infrastructure with 30+ regions and 90+ availability zones; elastic scalability that allows rapid scale in or out; strong security using the shared responsibility model with encryption and compliance certifications (SOC, ISO, HIPAA, PCI-DSS); high reliability through multi-AZ and multi-Region architectures; and the largest ecosystem and market leadership in cloud computing.

2. Explain the AWS Shared Responsibility Model.

The AWS Shared Responsibility Model defines how security responsibilities are divided between AWS and the customer. AWS is responsible for security OF the cloud, including physical data center security, hardware, networking, virtualization layer, and managed service infrastructure such as S3 durability or RDS engine patching. Customers are responsible for security IN the cloud, including data protection, IAM users and permissions, MFA, operating system patching for EC2 instances, network security configuration, application security, and encryption configuration. The shared responsibility varies by service type: in IaaS (EC2) customers manage more, while in managed services (Lambda, S3), AWS manages more. Understanding this model is critical for security and compliance.

3. What are AWS Regions, Availability Zones, and Edge Locations?

Regions are geographically isolated locations such as us-east-1 or ap-south-1, each independent with its own services and billing. Availability Zones are isolated data centers within a Region, connected via low-latency links, enabling high availability and fault tolerance. Edge Locations are global points of presence used by CloudFront and Route 53 to cache and deliver content closer to users for low latency. AWS also provides Local Zones for ultra-low latency and Wavelength Zones for 5G edge computing.

4. What is Amazon EC2 and what are its instance types?

Amazon EC2 provides resizable virtual machines in the cloud, allowing customers to choose operating systems, instance types, storage, and networking. Instance families include General Purpose (T, M) for balanced workloads; Compute Optimized (C) for CPU-intensive tasks; Memory Optimized (R, X) for in-memory databases; Storage Optimized (I, D) for high I/O workloads; and Accelerated Computing (P, G, Inf) for GPUs and machine learning. Purchasing options include On-Demand, Reserved Instances, Spot Instances, Savings Plans, and Dedicated Hosts.

5. What is Amazon S3 and what are its storage classes?

Amazon S3 is a scalable object storage service offering 99.999999999% durability. Data is stored as objects in buckets. Storage classes include Standard for frequent access, Intelligent-Tiering for unknown access patterns, Standard-IA for infrequent access, One Zone-IA for single-AZ storage, Glacier Instant Retrieval for archive access, Glacier Flexible Retrieval, and Glacier Deep Archive for long-term compliance storage. Key features include versioning, lifecycle policies, encryption, replication, and event notifications.

6. What is IAM in AWS?

AWS IAM controls authentication and authorization. Users are individual identities; groups organize users; roles provide temporary credentials for AWS services, cross-account access, or federated users; and policies define permissions in JSON format. Best practices include least privilege, MFA, avoiding root account use, and using roles instead of static access keys.

7. What is a VPC and how do you design a network in AWS?

A VPC is a logically isolated virtual network. It includes subnets within AZs, route tables, Internet Gateways, NAT Gateways, security groups (stateful), and network ACLs (stateless). Advanced networking includes VPC peering, Transit Gateway, and VPC Endpoints. Best practice design uses a multi-AZ architecture with public subnets for load balancers and private subnets for application and database tiers.

8. What is AWS Lambda and how does serverless computing work?

AWS Lambda is a serverless compute service that runs code in response to events without managing servers. Lambda automatically scales, manages capacity, and bills per millisecond of execution. It supports event sources such as API Gateway, S3, DynamoDB, and EventBridge. The maximum execution time is 15 minutes and concurrency can scale automatically. It is commonly used for APIs, event processing, automation, and microservices.

9. Differences between Security Groups and NACLs

Security Groups operate at the instance level, are stateful, and only allow rules. NACLs operate at the subnet level, are stateless, and support both allow and deny rules evaluated in order. Security Groups are the primary defense mechanism while NACLs add an additional subnet-level security layer.

10. What is Amazon RDS?

Amazon RDS is a managed relational database service supporting MySQL, PostgreSQL, MariaDB, SQL Server, Oracle, and Amazon Aurora. It provides automated backups, patching, Multi-AZ failover, read replicas, encryption, and performance monitoring.

11. What is DynamoDB?

DynamoDB is a fully managed NoSQL key-value and document database offering single-digit millisecond performance at scale. It supports on-demand and provisioned capacity, TTL, DAX caching, Streams, Global Tables, and transactions. It is ideal for large-scale, low-latency workloads.

12. Types of Load Balancers in AWS

AWS offers Application Load Balancers for HTTP/HTTPS, Network Load Balancers for TCP/UDP with ultra-low latency, and Gateway Load Balancers for traffic inspection appliances. These integrate with Auto Scaling and provide health checks and traffic distribution.

13. What is Auto Scaling?

AWS Auto Scaling dynamically adjusts resources based on demand using target tracking, step scaling, scheduled scaling, or predictive scaling. It integrates with ELB and CloudWatch to maintain desired performance and availability.

14. What is Amazon CloudFront?

Amazon CloudFront is a global content delivery network that caches content at edge locations, reducing latency and offloading origin servers. It supports HTTPS, Lambda@Edge, origin failover, signed URLs, and DDoS protection.

15. What is Route 53?

Amazon Route 53 is a scalable DNS and domain registration service. It supports routing policies such as simple, weighted, latency-based, failover, geolocation, geoproximity, and multi-value answers with built-in health checks.

16. What is AWS CloudFormation?

CloudFormation is an Infrastructure as Code service that uses YAML or JSON templates to provision AWS resources. It supports change sets, rollback, stack sets, nested stacks, and drift detection.

17. Difference between SQS and SNS

Amazon SQS is a message queue service for decoupling systems using a one-to-one model. SNS is a publish/subscribe service that delivers messages to multiple subscribers such as SQS, Lambda, HTTP endpoints, email, or SMS.

18. ECS vs EKS

Amazon ECS is an AWS-native container orchestration service that is simpler to operate. Amazon EKS is a managed Kubernetes service providing portability and Kubernetes ecosystem support. ECS suits AWS-centric workloads, while EKS suits Kubernetes-standard environments.